



Capstone project phase B

AI-Based Dorsiflexion Angle Measurement Using Foot Keypoint Detection

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by:

Sheli Zisman

Ron Kroitoro

Advisors: Dr. Naomi Unkelos Shpigel & Dr. Einat Ravid

Git repository link:

https://github.com/Sheli-Zisman/Dorsiflexion_FinalProject

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Abstract

Accurate assessment of ankle dorsiflexion is essential in rehabilitation and orthopedic practice, yet manual goniometric measurement remains prone to variability and limited reproducibility. This project developed an automated computer vision-based approach for estimating dorsiflexion angles directly from still images by detecting clinically relevant anatomical landmarks and applying vector-based angle computation.

A customized DeepLabCut model was trained on a dedicated dataset collected from real participants, and its performance was evaluated in collaboration with medical experts. The system provides a reliable and accessible tool for objective dorsiflexion assessment, with potential future applications in remote monitoring and digital clinical workflows.

1. Introduction

Clinical assessment in physical rehabilitation increasingly relies on quantitative measurements to support diagnosis, monitor recovery, and guide treatment. One important biomechanical parameter is the ankle dorsiflexion angle, which represents the range of motion between the foot and the shin. This measurement is relevant in the assessment of Achilles tendon injuries, neurological impairments, gait abnormalities, and musculoskeletal disorders. Traditionally, dorsiflexion is measured using a goniometer, but the result may be affected by examiner experience, positioning, and human error, limiting reliability and reproducibility [3].

Recent advances in computer vision and deep learning enable automated, image-based estimation of anatomical landmarks and joint angles [1], [5]. Such systems may reduce measurement variability, support non-invasive data collection, and provide more standardized biomechanical assessment [2]. DeepLabCut is especially suitable for this purpose because it allows customized detection of user-defined anatomical landmarks using relatively small labeled datasets and deep neural networks such as ResNet50 [5].

Despite this progress, the clinical use of computer-vision-based ankle dorsiflexion measurement remains limited. Existing solutions often focus on general human movement rather than task-specific static ankle assessment. In addition, challenges related to accuracy, standardization, and clinical implementation remain [8].

This project developed a computer vision system for estimating ankle dorsiflexion from still images. The system detected landmarks on the shin, ankle, and fifth metatarsal and calculated the angle using a vector-based method. The project included participant data collection, model training, algorithm comparison, and accuracy evaluation. In collaboration with medical experts at Hospital X, the preliminary results were reviewed, and their feedback was used to refine the system and improve its clinical relevance. The project provides an accessible tool that may support clinical assessment and future remote rehabilitation applications.

The system is based on the integration of DeepLabCut and a ResNet-50 neural-network backbone. This combination was selected because it provides the flexibility and precision required for detecting small, clinically meaningful anatomical landmarks in images of the foot and ankle[5],[8].

DeepLabCut is a markerless pose-estimation framework designed for detecting user-defined keypoints in images and videos. Unlike general-purpose pose-estimation systems that rely on a fixed set of full-body landmarks, DeepLabCut allows the development team to define the exact anatomical points required for the task[1],[5]. In this project, these points included landmarks located on the shin, ankle, and fifth metatarsal, which are later used to calculate the ankle dorsiflexion angle.

The framework follows a supervised-learning approach. A representative subset of the collected images is manually annotated by marking the required anatomical landmarks. These labeled images were then used to train the model to recognize the same points in previously unseen images. During prediction, DeepLabCut assigns image coordinates to each detected landmark, together with a confidence score that represents the reliability of the prediction[5]. Predictions with low confidence can be identified and reviewed, which supports quality control during model development and evaluation.

DeepLabCut is particularly suitable for this project because the required landmarks are small and clinically specific. General pose-estimation models are commonly trained to locate major joints, such as the shoulders, hips, knees, and ankles, but they are not designed to identify detailed structures around the foot. By using a task-specific annotated dataset, DeepLabCut can learn the visual characteristics of the selected landmarks and adapt to the particular image conditions of the project[5],[8].

ResNet-50 is used as the feature-extraction backbone of the DeepLabCut model [5]. It is a deep convolutional neural network composed of 50 layers and is designed to identify visual patterns at different levels of abstraction. Early network layers detect basic features such as edges, contours, and textures, while deeper layers learn more complex structures related to the shape and position of the foot and ankle.

ResNet-50 uses residual connections that improve the training stability of deep neural networks and enable the extraction of complex visual features from medical images. The network is initialized using transfer learning, starting from pre-trained weights and then fine-tuned on the project's annotated foot and ankle dataset, reducing data requirements and improving prediction accuracy [5].

After landmark detection, the two-dimensional coordinates are used to construct vectors representing the shin and foot orientations. The dorsiflexion angle is then calculated from the geometric relationship between these vectors.

The combination of DeepLabCut and ResNet-50 supports markerless detection of clinically relevant landmarks using a relatively small task-specific dataset and provides the ability to learn visual variations in anatomy and image-acquisition conditions [4], [5].

2. Background

2.1 Clinical Background

2.1.1 Clinical Importance of Ankle Dorsiflexion Measurement

Ankle dorsiflexion is the movement that decreases the angle between the foot and the tibia. It is essential for normal gait, balance, and functional mobility, particularly during movements in which the body advances over the supporting foot. Restricted dorsiflexion can alter lower-limb biomechanics and may contribute to abnormal walking patterns and reduced functional performance [6].

Dorsiflexion measurement is commonly used when assessing patients with Achilles tendon injuries, ankle sprains, neurological impairments, and following surgical procedures. Accurate measurement allows clinicians to evaluate the severity of movement limitations, monitor recovery over time, and adjust rehabilitation programs according to the patient's progress.

Because clinical decisions may depend on relatively small changes in joint angle, the measurement method should be consistent and reproducible. However, conventional goniometric assessment may vary according to examiner technique and anatomical landmark placement [3]. This creates a clinical need for an objective measurement approach that can support reliable follow-up and comparison between examinations.

2.1.2 Current Method for Measuring Ankle Dorsiflexion

In current clinical practice, ankle dorsiflexion is commonly measured manually using a goniometer. The patient is typically positioned in a seated or supine position, with the knee maintained according to the selected examination protocol. In some measurements, the knee is slightly flexed to reduce the influence of the gastrocnemius muscle on ankle movement.

Before placing the goniometer, the clinician identifies the relevant anatomical landmarks through visual inspection and palpation. The lateral malleolus is commonly used as the axis of rotation for the ankle joint. The fulcrum of the goniometer is placed over this point, while its stationary arm is aligned with the longitudinal axis of the tibia. The moving arm is positioned along the lateral border of the foot, usually following the direction of the fifth metatarsal [7].

The patient is then instructed to move the foot into maximum active dorsiflexion. Alternatively, the clinician may perform a passive measurement by moving the foot while attempting to maintain the required alignment. Once the maximum position is reached, the angle is read directly from the goniometer scale and recorded as the ankle dorsiflexion range of motion.

Although the procedure is relatively simple and requires only basic equipment, its accuracy depends on several manual steps. The clinician must correctly identify the anatomical landmarks, position the patient consistently, align both arms of the goniometer, maintain this alignment during movement, and visually interpret the final value. Small differences in any of these stages may influence the resulting angle.

For example, inaccurate placement of the fulcrum over the lateral malleolus may change the assumed joint axis, while slight deviations in the tibial or foot alignment may alter the measured angle. Differences in knee position, applied force during passive movement, or the patient's effort during active movement may also affect the result. Consequently, measurements may vary both between different clinicians and between repeated assessments performed by the same clinician [3].

2.2 Literature Review

2.2.1 Computer-Vision Tools for Clinical Measurement

Computer vision has become an important tool for quantitative clinical and biomechanical assessment. Deep-learning models can detect anatomical landmarks in images and use their locations to estimate joint positions and clinically relevant angles [2], [8].

Unlike traditional motion-capture systems, modern computer-vision approaches can operate using standard cameras without requiring markers, wearable sensors, or specialized laboratory equipment [2], [8]. This enables more accessible and standardized assessment for applications such as gait analysis, range-of-motion evaluation, and rehabilitation monitoring [2],[8].

These systems commonly rely on deep neural networks to extract visual features and estimate anatomical landmark coordinates, which can then be converted into quantitative biomechanical measurements such as joint angles and movement parameters.

2.2.2 Markerless Pose Estimation

Markerless pose estimation identifies anatomical landmarks directly from images without requiring physical markers or wearable sensors. By using neural networks to estimate landmark coordinates, these methods provide a less intrusive and more accessible alternative to traditional motion-capture systems [8].

Pose-estimation approaches can be divided into general-purpose and task-specific models. General-purpose systems such as OpenPose detect predefined body landmarks and are well suited for full-body analysis [1]. However, they may lack the anatomical detail required for clinically specific regions such as the foot and ankle.

Task-specific frameworks such as DeepLabCut allow researchers to define custom anatomical landmarks and train models on dedicated annotated datasets [5]. Combined with transfer

learning, this approach reduces data requirements while enabling accurate detection of clinically relevant landmarks for ankle dorsiflexion assessment [5].

2.2.3 Pose Estimation and Deep Learning in Clinical Biomechanics

Deep-learning-based pose estimation has expanded the possibilities of clinical biomechanics by enabling quantitative movement analysis from standard camera recordings. Kidziński et al. demonstrated that clinically relevant movement measurements can be extracted from single-camera videos without requiring complex laboratory equipment [2].

In rehabilitation and movement analysis, detected landmarks can be used to calculate joint angles, evaluate posture, and monitor functional changes over time. However, clinically meaningful assessment requires accurate landmark definitions, standardized image acquisition, and validation against accepted reference measurements [8].

These considerations are particularly important for ankle dorsiflexion, where small localization errors may substantially affect the calculated range of motion. Therefore, a task-specific model trained on foot and ankle images is more suitable than a general full-body pose estimator.

The literature supports the use of DeepLabCut with a deep convolutional backbone for customized anatomical landmark detection from standard images [5]. In this project, these principles were applied to estimate ankle dorsiflexion ROM by calculating the difference between the ankle angles measured in the resting and dorsiflexion images.

3. Engineering Process

3.1 User-Centered Research

The system was developed according to clinical requirements for ankle range-of-motion assessment. Throughout the project, interviews and consultations with orthopedic specialists from Hospital X, including senior clinicians involved in foot and ankle assessment, guided the selection of anatomical landmarks, participant positioning, and measurement procedures, and helped define the clinical requirements of the system.

The intended users of the system are orthopedic physicians, physiotherapists, and other rehabilitation professionals involved in ankle assessment and recovery monitoring.

To ensure accessibility and ease of use, the system relies on standard images captured with a regular camera, without requiring markers, wearable sensors, or specialized laboratory equipment.

In addition, surveys were conducted among orthopedic physicians, physiotherapists, and rehabilitation professionals from a variety of clinical settings and institutions. The responses highlighted limitations of current assessment methods, identified desirable

features for future clinical tools, and provided feedback regarding the usefulness and potential applicability of the developed system.

This user-centered approach ensured that the system addressed not only prediction accuracy, but also clinical relevance, usability, and measurement consistency.

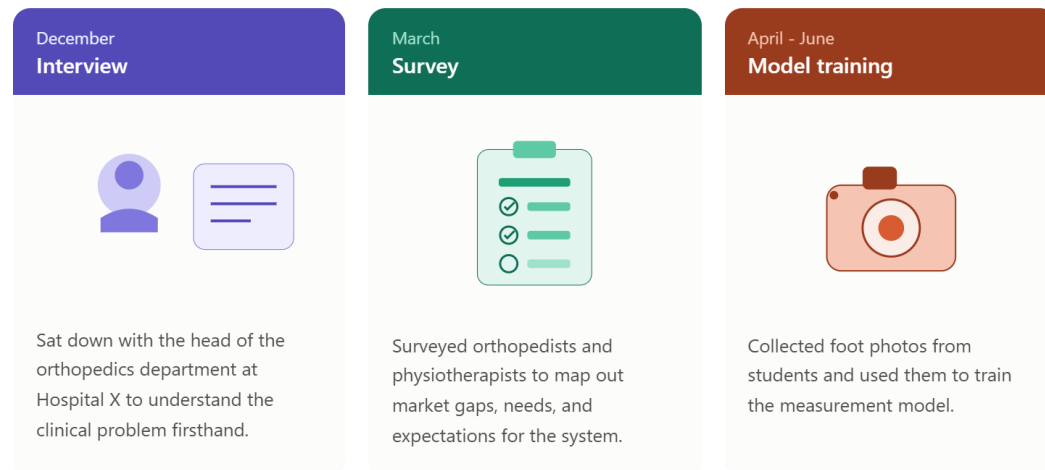


Figure 1. TIMELINE

3.2 Project Workflow: Automated Ankle Measurement System

Figure 2 presents an overview of the project workflow for automated ankle dorsiflexion measurement in three main phases. It begins with the clinical motivation and the limitations of manual goniometric assessment, followed by the evaluation of alternative pose-estimation approaches and the selection of a customized DeepLabCut-based solution. Finally, the diagram illustrates the operational pipeline in which the user uploads an image, anatomical landmarks are detected automatically, and the dorsiflexion angle is computed and displayed.

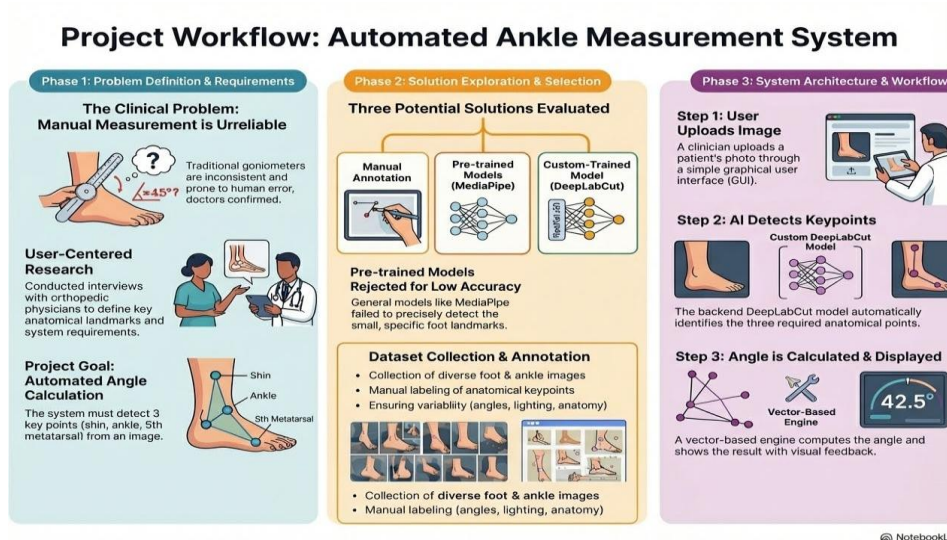


Figure 2: System Workflow for Automated Ankle Dorsiflexion Measurement

The project began with defining the clinical problem and collecting requirements in collaboration with a senior orthopedic physician specializing in foot and ankle disorders. Several pose-estimation approaches were evaluated to identify the most suitable solution for accurate anatomical landmark detection.

The final system automatically detects three anatomical landmarks: the tibial axis, lateral malleolus, and fifth metatarsal. Using these points, ankle dorsiflexion ROM is calculated by comparing images captured in neutral and maximal dorsiflexion positions.

The development process included dataset collection, image annotation, model training, performance evaluation, and iterative refinement to achieve clinically meaningful accuracy and reliability.

Figure 3 highlights the motivation for transitioning from traditional manual ankle measurement methods to AI-based precision tools. It summarizes the key limitations of goniometric assessment, including human error, lack of standardization, and inconsistent landmark identification. In contrast, the proposed solution leverages computer vision and deep learning to enable automated, accurate, and clinically reliable dorsiflexion angle estimation from images.

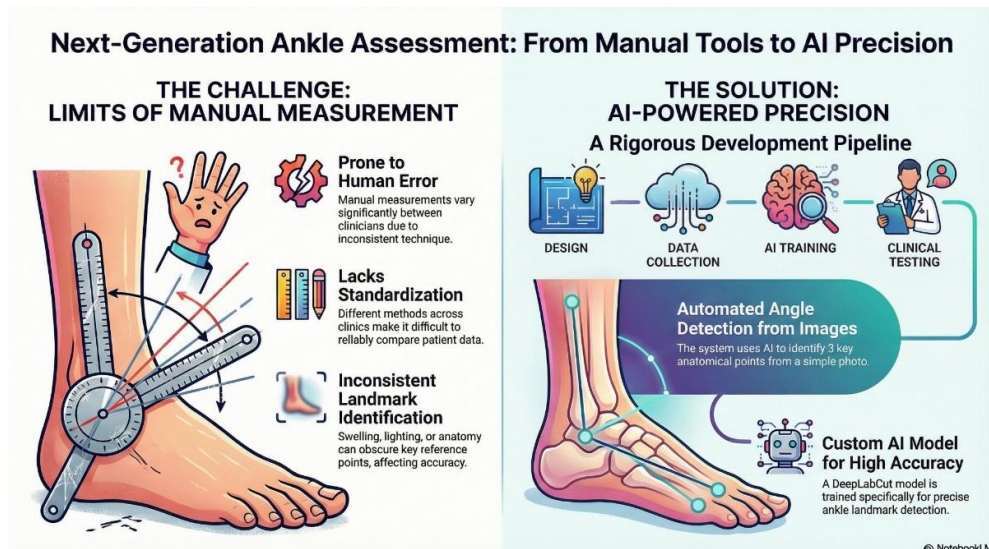


Figure 3: Next-Generation Ankle Assessment

3.3 Key Insights from Consultations with Medical Professionals

The interviews focused on the limitations of current dorsiflexion measurement methods, the selection of clinically meaningful anatomical landmarks, and the requirements for a practical image-based measurement procedure. The main insights are:

1. Variability in Manual Goniometric Measurement

The medical professionals emphasized that manual goniometric measurements may vary according to the examiner's experience, positioning technique, and ability to identify the relevant anatomical landmarks. Small differences in the placement of the goniometer fulcrum or in the alignment of its arms with the tibia and foot may affect the resulting angle. This variability can make it difficult to compare measurements performed by different clinicians or to reliably monitor small changes during rehabilitation.

2. Importance of Anatomical Landmark Selection

A major part of the consultation concerned the anatomical points required for calculating ankle dorsiflexion. The selected landmarks needed to represent the direction of the shin and the foot while remaining identifiable in lateral images. The medical guidance contributed to the selection and placement of three landmarks used by the system and to ensuring that the calculated angle corresponded to the clinical range-of-motion measurement.

The discussion also highlighted that landmark visibility may differ between participants because of anatomical variation, calf-muscle structure, body composition, swelling, skin appearance, and foot shape. These factors reinforced the need for both clinically valid landmark definitions and a model trained on images representing different participants.

3. Need for a Standardized Measurement Protocol

The consultations showed that reliable automated measurement depends not only on the machine-learning model but also on the way the images are captured. The participant's knee angle, chair position, camera height, camera distance, and viewing angle must be controlled as consistently as possible. In particular, maintaining the knee at approximately 90 degrees was considered important for creating comparable measurements.

This requirement presented a practical challenge because participants differed in height and body proportions. Therefore, the experimental setup had to balance a fixed camera configuration with adjustments needed to preserve the required clinical posture.

3.4 Requirements

The system requirements were defined based on clinical consultations, the objectives of the project, and the technical constraints identified during development. The clinically oriented requirements, including landmark selection and comparison with goniometric measurements, were determined with guidance from medical professionals at Hospital X. The technical and performance requirements were defined by the development team in order to support accurate, consistent, and practical system operation.

3.4.1 Functional Requirements

1. The system shall receive a still image containing a lateral view of the participant's lower leg and foot.
2. The system shall detect three clinically selected anatomical landmarks representing the shin, ankle region, and fifth metatarsal.
3. The system shall return the two-dimensional coordinates of each detected landmark.
4. The system shall assign a confidence value to each detected landmark.
5. The system shall identify cases in which one or more landmarks are detected with insufficient confidence.
6. The system shall calculate the ankle dorsiflexion angle using vectors constructed from the three detected landmarks.
7. The system shall display the calculated dorsiflexion angle in a clear numerical format.
8. The system shall support analysis of images captured in both resting and dorsiflexion positions.
9. The system shall allow the calculated result to be compared with a manual goniometric measurement collected from the same participant.
10. The system shall preserve the association between each image, its annotated landmarks, its predicted landmarks, and its calculated angle for evaluation purposes.

3.4.2 Non-Functional Requirements

Accuracy: The system should achieve an absolute dorsiflexion-angle error of no more than 5 degrees when compared with the corresponding clinical goniometric measurement.

Performance: The system should process a single image and produce the detected landmarks and calculated angle within 10 seconds under the selected execution environment.

Reliability: The system should consistently detect all three required landmarks in images captured according to the standardized acquisition protocol.

Consistency: Images used by the system should be captured under controlled conditions, including a fixed camera position, consistent chair placement, a white background, and an approximately 90-degree knee position.

Usability: The system output should present the predicted landmarks, confidence values, and calculated angle in a form that can be understood by the development team and medical professionals.

Privacy: Participant images shall be collected only after informed consent has been obtained. Personal identifiers shall not be included in the image filenames, annotations, or evaluation results.

Maintainability: The landmark-detection component and the angle-calculation component shall be implemented as separate modules so that each can be updated independently.

Portability: The trained model and angle-calculation process should be executable on standard computing environments without requiring specialized motion-capture hardware.

Reproducibility: The same trained model and calculation method shall be applied consistently to all evaluation images.

Extensibility: The system architecture should allow future integration with a graphical interface, mobile-device image capture, cloud-based execution, and long-term storage of measurement results.

3.5 Solution

The solution is an image-based application for estimating ankle dorsiflexion range of motion. The application receives two lateral images of the same leg: one image in the resting position and one image in maximum dorsiflexion. In both images, the participant remains seated with the knee flexed at approximately 90 degrees and the lower leg positioned vertically relative to the floor. The same participant position, camera angle, and image-acquisition conditions are maintained for both images.

The solution combines a customized DeepLabCut model with a ResNet-50 backbone and a geometric angle-calculation component. Its workflow consists of image acquisition, automatic anatomical landmark detection, angle calculation for each image, and computation of the final range-of-motion value.

Image Acquisition

The user captures two images of the same leg from a lateral view. The first image represents the resting position of the foot, while the second represents the maximum dorsiflexion position. To improve consistency, the participant remains seated in the same position during both captures, with the knee at approximately 90 degrees and the lower leg vertical to the floor.

During the experimental stage of the project, the camera was mounted on a tripod at a fixed location, the chair was positioned consistently, and a white background was placed behind the participant's leg. These conditions were selected to reduce unnecessary variation between images and to improve the visibility of the leg and foot.

Anatomical Landmark Detection

For each image, the system detects three clinically selected anatomical landmarks representing the shin, ankle region, and fifth metatarsal. These landmarks were selected in collaboration with a senior orthopedic specialist to ensure consistency with conventional dorsiflexion assessment.

Landmark detection is performed using DeepLabCut, a markerless pose-estimation framework, with ResNet-50 serving as the feature-extraction backbone. ResNet-50 extracts visual features from the image, while DeepLabCut uses these features to localize the predefined anatomical landmarks and estimate their coordinates.

The model was trained on manually annotated images and returns the two-dimensional coordinates of each landmark together with a confidence score. These confidence values help identify unreliable predictions caused by factors such as poor visibility, lighting conditions, anatomical variation, or partial occlusion.

Angle Calculation

After the three landmarks are detected, the system constructs two vectors. One vector represents the orientation of the lower leg, while the second represents the orientation of the foot. The angle between these vectors is calculated separately for the resting image and for the dorsiflexion image.

The ankle range of motion is calculated by subtracting the angle measured in the resting image from the angle measured in the dorsiflexion image. The absolute value of this difference is used as the final ROM result.

Using two images allows the application to account for the participant's initial resting position instead of relying only on a single absolute ankle angle.

Result Presentation

After both images have been analyzed, the application displays the final ROM value in degrees. The predicted anatomical landmarks may also be displayed on the original images, allowing the user to visually verify that the model identified the intended locations.

If the confidence level of one or more detected landmarks is below the required threshold, the application can notify the user that the measurement may be unreliable and that the image should be captured again.

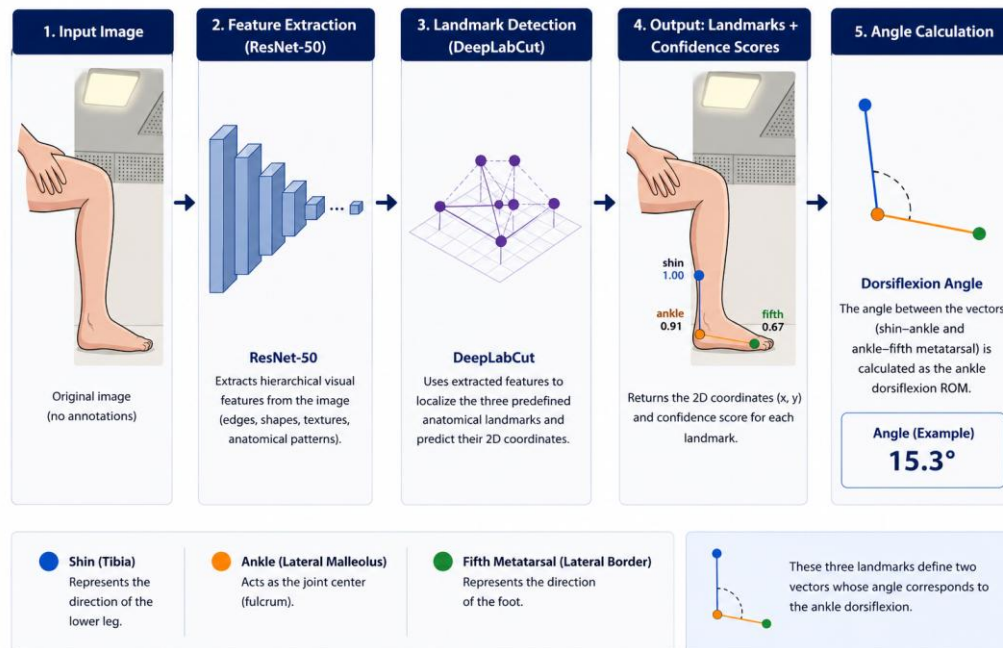


Figure 4 :DeepLabCut-ResNet50 Landmark Detection Pipeline

Clinical Validation

As a future stage of the project, the ROM values produced by the application will be compared with manual goniometric measurements collected from the same participants.

This validation process will be used to evaluate the accuracy and clinical reliability of the system, since accurate landmark detection alone does not guarantee agreement with conventional clinical measurements. Comparison with goniometric assessment will therefore provide an important reference for determining the suitability of the system for future clinical use.

4 System Architecture and Design

4.1 System Architecture Overview

The system is designed as a modular pipeline that converts two clinical images into an automated ankle dorsiflexion ROM measurement. The user uploads images captured in neutral and maximal dorsiflexion positions under standardized conditions.

The images are processed by a backend service and analyzed using a DeepLabCut model with a ResNet-50 backbone, which automatically detects three anatomical landmarks: the tibial axis, lateral malleolus, and fifth metatarsal.

Using these landmarks, the system calculates the ankle angle in each image through vector-based geometric analysis and determines ROM as the difference between the two measurements.

The final result, along with the detected landmarks and calculated vectors, is presented to the user to support transparency and clinical validation.

4.2 Technology Stack Overview

The current system is implemented using a focused technology stack that supports image-based landmark detection, model training, and ankle range-of-motion calculation.

Python 3.x is used as the primary development language for the machine-learning pipeline, image processing, geometric calculations, and evaluation procedures.

DeepLabCut is used as the main framework for markerless anatomical keypoint detection. It enables the model to be trained on manually annotated images and to predict the locations of the three clinically selected landmarks in previously unseen images.

ResNet-50 is used as the feature-extraction backbone within the DeepLabCut model. Its deep residual architecture enables the model to learn visual characteristics of the lower leg, ankle, and foot and supports accurate landmark localization.

OpenCV is used for image-related operations, including image loading, resizing, visualization, and preparation of images for analysis.

NumPy is used for numerical processing and vector-based calculations. The coordinates predicted by the model are converted into vectors, which are then used to calculate the ankle angle in each image and the final range-of-motion value.

The current technology stack focuses on the core processing pipeline: image input, landmark prediction, angle calculation, and result evaluation. The system is designed in a modular manner so that the machine-learning component, image-processing functions, and ROM calculation can be developed and tested independently.

Future development may include integration with a web or mobile interface, a backend API, and local or cloud-based storage. These additions would allow users to capture or upload the

two required images, receive the calculated ROM value, and store previous measurements for later comparison.

Architecture diagram :

Figure 5 presents the high-level architecture of the proposed dorsiflexion measurement system, showing the interaction between the user interface, backend server, and the DeepLabCut-based inference module.

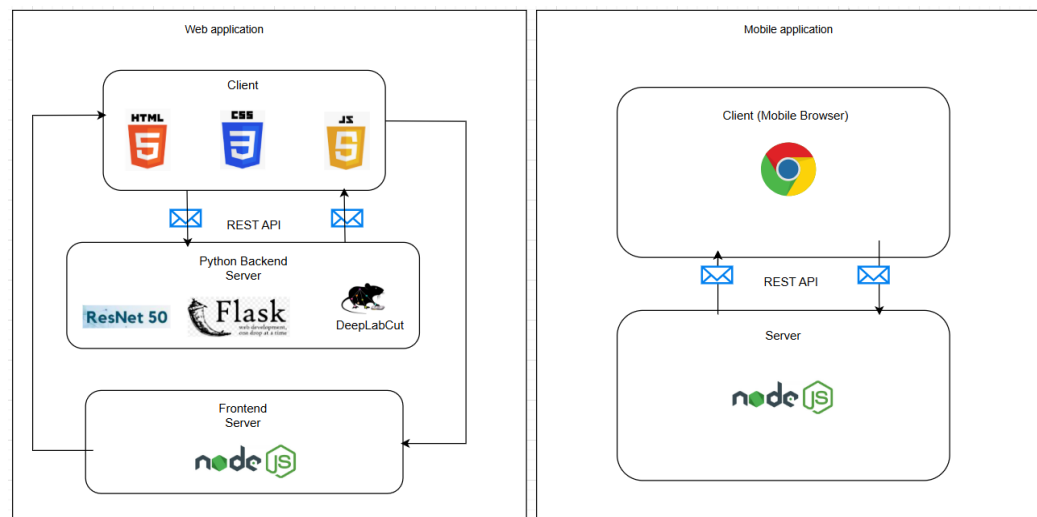


Figure 5:Architecture diagram

4.3 System Components and Technology Overview

The system is composed of several modular components that work together to transform an input image into a clinically meaningful dorsiflexion measurement. Each component fulfills a specific role within the processing pipeline, allowing the system to operate efficiently, ensure accuracy, and maintain flexibility for future expansion. The following subsections describe the main components and the technologies that support them.

4.3.1 User Interface Layer

The graphical user interface is designed to provide a simple and intuitive workflow for ankle range-of-motion measurement. The main screen contains two image panels: one for the resting position and one for maximum dorsiflexion.

Users can upload or capture an image in each panel and preview it before analysis. Images can be removed or replaced if necessary. Short instructions guide the user to capture both images from a lateral view while maintaining the same seated posture, camera position, and knee angle.

After both images are provided, the user can start the analysis. The interface sends the images for landmark detection and ROM calculation, then displays the final result in degrees. If the

model detects one or more landmarks with low confidence, the interface notifies the user that the relevant image should be replaced.

4.3.2 Backend Processing Layer

The backend layer coordinates the main processing workflow. It receives the resting and dorsiflexion images, applies the required preprocessing steps, and sends both images to the machine-learning inference module.

After the three anatomical landmarks are detected in each image, the backend calculates the corresponding ankle angles and determines the final ROM value from the difference between them. It then returns the result to the user interface for display.

The backend is designed in a modular structure, allowing the preprocessing, landmark-detection, and angle-calculation components to be updated independently.

4.3.3 Machine Learning Inference Module

The machine-learning inference module identifies the three anatomical landmarks in both the resting and dorsiflexion images. It is implemented using DeepLabCut and trained on a customized dataset of manually annotated foot and ankle images collected during the project.

A ResNet-50 backbone is used for feature extraction and supports accurate localization of the selected shin, ankle, and fifth-metatarsal landmarks. For each image, the module returns the coordinates of the detected points together with confidence values that indicate prediction certainty.

Because the inference module is separated from the remaining system components, the model can be retrained or replaced without changing the backend or user interface.

4.3.4 Angle-Computation Engine

After receiving the predicted keypoints, the angle-computation engine constructs vectors representing the lower leg and foot in each image. It then calculates the ankle angle separately for the resting and dorsiflexion positions using vector-based geometric methods.

The final ROM value is obtained from the absolute difference between the two calculated angles. This method is consistent, transparent, and easy to interpret, making it suitable for clinical evaluation and future system extensions.

4.3.5 Future Data Storage and Logging

A future version of the system may include a storage module for saving ROM results, timestamps, and model confidence values. This would allow measurements to be reviewed over time and support longitudinal monitoring of patient progress.

The stored data could also support future integration with electronic medical record systems and may be used, subject to appropriate consent and privacy controls, for further model evaluation and retraining.

4.4 Pilot Experiment

4.4.1 Experimental Procedure

The experiment was conducted to evaluate the accuracy, reliability, and processing time of the proposed automated ankle dorsiflexion measurement system in comparison with manual goniometric assessment.

Each measurement session consisted of two lateral images of the same participant: one image with the foot in a resting position and one image in maximum dorsiflexion. In both images, the participant remained seated with the knee flexed at approximately 90 degrees and the lower leg positioned vertically relative to the floor.

Before participating, each student signed an informed-consent form approving the collection and use of the images for the project.

The experimental procedure included the following stages:

Image Acquisition: Two images of the lower leg and foot were captured for each participant. A fixed tripod position, consistent chair placement, standard indoor lighting, and a white background were used to improve image consistency and landmark visibility.

Data Preparation: The collected images were organized into measurement sessions and manually annotated using the three selected anatomical landmarks. The annotations were used for model training and evaluation.

Automated Keypoint Detection: The trained DeepLabCut model detected the shin, ankle, and fifth-metatarsal landmarks in both images and returned their coordinates and confidence values.

Angle and ROM Computation: The system calculated the ankle angle separately for the resting and dorsiflexion images. The final ROM value was obtained from the absolute difference between the two calculated angles.

Result Recording: The automated ROM value, landmark predictions and confidence values were recorded for subsequent evaluation.

4.4.2 Participant Characteristics

The experiment included healthy adult volunteers primarily undergraduate students, between the ages of 20 and 60. Participants reported no active ankle injury or known musculoskeletal condition that could significantly affect ankle movement at the time of data collection.

The participant group included individuals with different heights, body structures, and calf-muscle characteristics. This variation was important for evaluating the model across different anatomical appearances and also introduced practical challenges in maintaining consistent positioning and landmark visibility.

All participants received an explanation of the data-collection procedure and signed an informed-consent form before their images and measurements were collected.

4.4.3 Evaluation Criteria

The system was evaluated using three main criteria: keypoint-detection reliability, processing time, and measurement accuracy. While the first two were evaluated in the present study, measurement accuracy will be assessed in future clinical validation.

Measurement Accuracy: The target is to achieve a mean absolute error of no more than 5 degrees during future clinical validation.

Detection Reliability: The target was to successfully detect all three required anatomical landmarks in at least 95% of the evaluated measurement sessions captured under the standardized experimental conditions.

Processing Time: The target was to complete the analysis of both the resting and dorsiflexion images and produce the final ROM value within 10 seconds.

These criteria were used to determine whether the system achieved the accuracy, reliability, and efficiency requirements defined for the project.

4.4.4 Data Processing

All collected images were processed using the developed system. Each measurement session included a resting image and a maximum-dorsiflexion image of the same participant.

The images were analyzed using a DeepLabCut model with a ResNet-50 backbone, which extracted visual features and detected the three predefined anatomical landmarks in both images. The system then calculated the ankle angle for each image and determined the final ROM value from the difference between the two angles.

The predicted landmarks were visualized on the images to support verification of the model's detections. The automated ROM values, confidence scores, and processing times were recorded for comparison and further analysis.

4.4.5 Evaluation Metrics

System performance was evaluated using quantitative metrics corresponding to the project requirements.

Keypoint Detection Reliability: Detection reliability was measured as the percentage of measurement sessions in which all required anatomical landmarks were successfully detected in both images.

Processing Time: Processing time was defined as the total time required to process the images, detect the landmarks, calculate the ankle angles, and generate the final ROM value.

Prediction Confidence: The confidence scores produced by DeepLabCut were recorded for each landmark to identify uncertain detections and evaluate prediction reliability.

Robustness to Anatomical Variation: Robustness was evaluated by examining model performance across participants with different body structures and anatomical characteristics.

5. Results

Table 1. DeepLabCut model evaluation

Metric	Train	Test
RMSE (pixels, all predictions)	6.07	8.05
RMSE (pixels, at p-cutoff)	5.27	7.52
mAP (%)	97.54	94.48
mAR (%)	98.41	96.76

The model achieved strong landmark detection performance, demonstrating good generalization and reliable localization of the anatomical points required for dorsiflexion assessment.

However, the current evaluation is limited to landmark detection metrics and does not yet include direct comparison with manual goniometric measurements.

Future work will focus on clinical validation of the calculated ROM values to assess the accuracy and applicability of the system in practice.

Table 2: Time Estimates for Each Stage of the Measurement Process

Estimated Time	Description	Stage	Actual results
~1 minute	Capture of two images per participant	Image acquisition	~1 minute
~15 seconds	Upload images to the system	Data upload	~9 seconds
~5 seconds	Automated Processing Pipeline	Keypoint detection, angle calculation and Data recording	~12 seconds

The evaluation showed that some stages of the workflow performed better than initially expected, particularly the image upload process, which proved to be efficient and user-friendly. Image acquisition times were consistent with the original expectations. In contrast, the automated processing stage required more time than anticipated, indicating that model inference and computation remain the main areas for future optimization. Despite this limitation, the overall workflow remained practical and suitable for routine use.

6. Discussion and conclusions

6.1 Discussion

The project successfully demonstrated the feasibility of using computer vision for automated ankle dorsiflexion assessment from standard images. The DeepLabCut-ResNet50 model achieved strong landmark detection performance while requiring only a standard camera and no additional sensors, markers, or specialized equipment [2], [5]. However, the current evaluation focused primarily on landmark detection accuracy and did not yet include direct

comparison with manual goniometric measurements. In addition, system performance may still be influenced by image acquisition conditions such as lighting, participant positioning, and anatomical variation [8]. Future work will therefore focus on clinical validation against goniometric assessment, expansion of the training dataset, and optimization of the processing pipeline to improve robustness and applicability in clinical environments.

6.2 Conclusions

The application integrates controlled image acquisition, customized anatomical landmark detection, and automated geometric calculation into a single measurement process. Its purpose is to provide a clear and repeatable estimate of ankle dorsiflexion ROM while reducing dependence on manual landmark alignment and visual angle reading.

The current project focuses on the development, training, and validation of the detection and calculation components. Future development may include improved user guidance during image capture, automatic posture verification, result storage, and longitudinal comparison of measurements over time.

7.Challenges Encountered

The development of the automated dorsiflexion measurement system involved several practical, clinical, and technical challenges. Recruiting suitable participants required additional coordination, as the study excluded individuals with active ankle conditions and relied on informed consent and structured measurement sessions.

Maintaining a standardized image-acquisition setup also proved challenging due to differences in participant height, body proportions, and anatomy. Variations in foot shape, calf-muscle structure, and soft-tissue distribution affected the visibility and consistency of the selected anatomical landmarks, creating difficulties during both manual annotation and automatic prediction.

Environmental factors such as lighting conditions, shadows, camera angle, and image background also influenced landmark visibility and model performance. In addition, continuous clinical guidance was required for landmark selection and measurement protocol design, although scheduling consultations was occasionally difficult.

Finally, the planned clinical validation introduces additional complexity, since manual goniometric measurements themselves contain examiner-dependent variability. These challenges highlighted the importance of standardized image acquisition, consistent annotation procedures, and the need for larger and more diverse datasets in future work.

Appendices

8. AI Tools

Google Gemini: <https://gemini.google.com/>

It was used as an auxiliary AI tool to assist in generating and refining key visual diagrams for the project, including the project workflow overview (Figure 1), the next-generation ankle assessment illustration (Figure 2), and the system architecture schematic (Figure 3) presented in this book.

ChatGPT (OpenAI): <https://chat.openai.com/>

Was used as an AI-based research and writing assistant to support literature understanding, academic phrasing, and clarification of technical concepts throughout the project.

9. Use Case Diagram

Ankle Dorsiflexion Use Cases:

Figure A1 presents the main interactions between the user and the automated ankle dorsiflexion measurement system. The primary process involves uploading two foot images: initial and dorsiflexion, after which the system automatically detects anatomical landmarks, performs angle calculation, and generates ROM visualization. These steps are modeled as included use cases since they are essential for the measurement workflow. An additional optional use case allows comparison with manual goniometric measurements to support clinical validation. Overall, the diagram clarifies the system boundary and the division between user actions and automated processing.

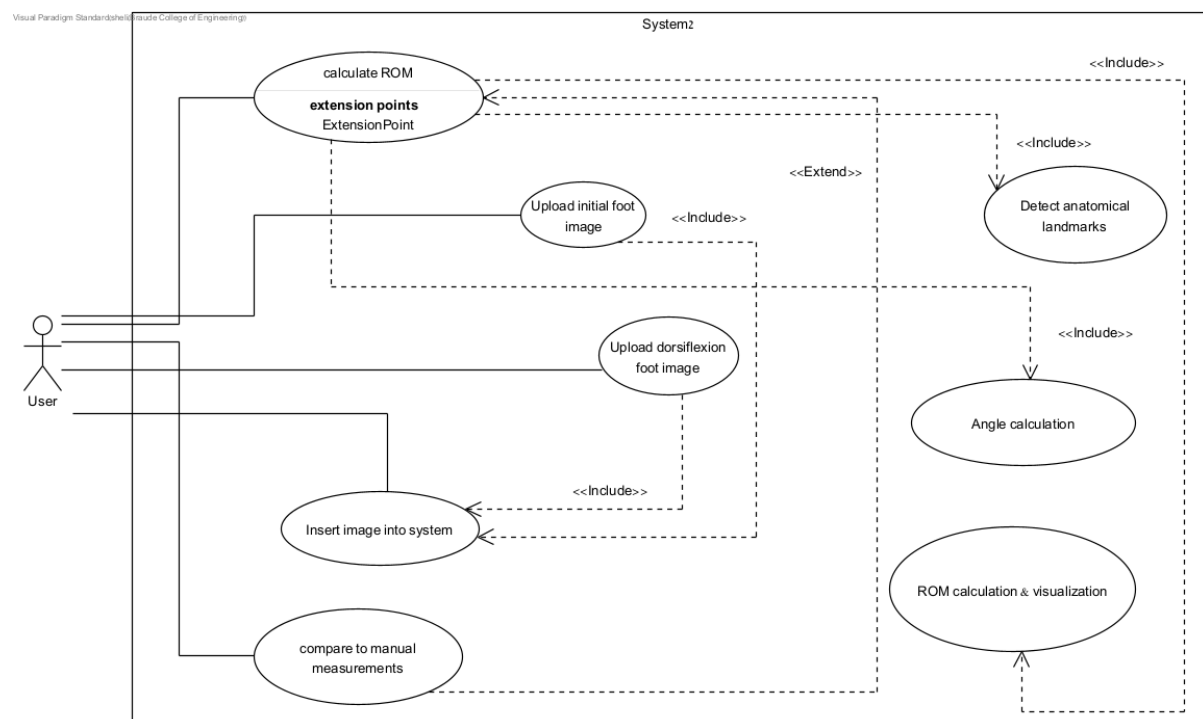


Figure A1:Ankle Dorsiflexion Use Cases

10. Activity Diagram

Figure A2 illustrates the workflow of the automated dorsiflexion measurement process, divided between user actions and system processing. The user allows camera access and uploads the required foot images, while decision points ensure the inputs are suitable for analysis. The system then performs landmark detection, computes foot vectors, and assesses the dorsiflexion ROM. Finally, the result is returned to the user for visualization, completing the session.

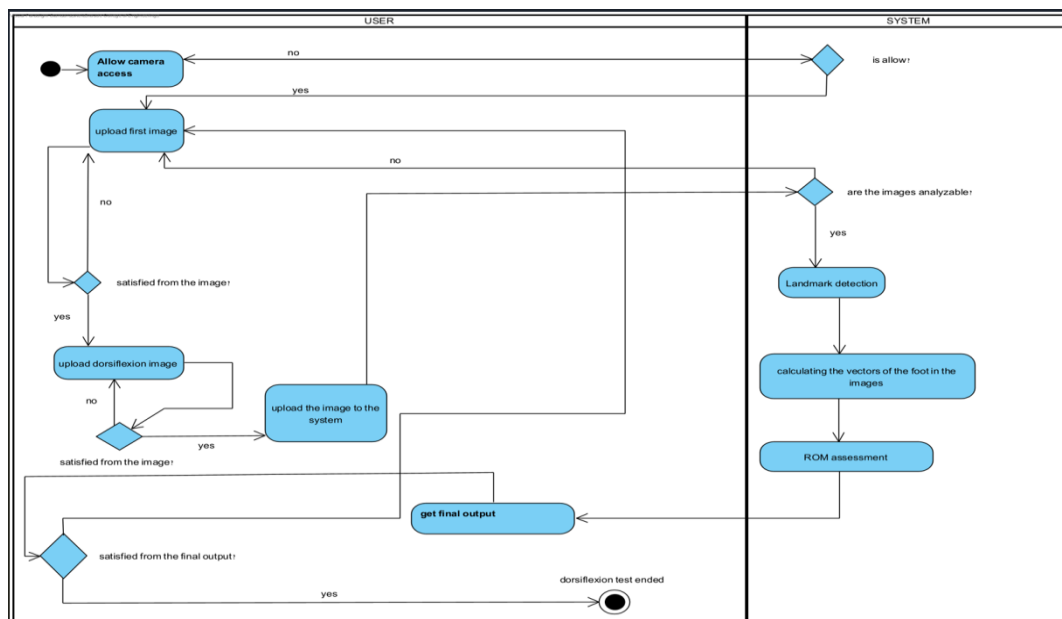


Figure A2: Ankle Dorsiflexion Measurement Activity

User Guide

1. System Purpose

The system is designed to calculate ankle dorsiflexion range of motion using two images: one image in the resting position and one image in maximum dorsiflexion.

The system automatically detects three anatomical landmarks and calculates the angular difference between the two positions.

2. Image-Capture Requirements

Before using the system, the images should be captured under appropriate conditions:

- The participant should sit on a chair.
- The knee should be flexed at approximately 90°.
- The lower leg should remain as vertical as possible.
- The image should be captured from the side.
- The lower leg, ankle, and foot should be fully visible.
- Both images should be captured from the same distance and camera angle.
- Good lighting and a clean background should be used.

3. Opening the Application

The application is opened through a web browser.

The main screen contains two image-upload areas and a button for calculating the ankle range of motion.

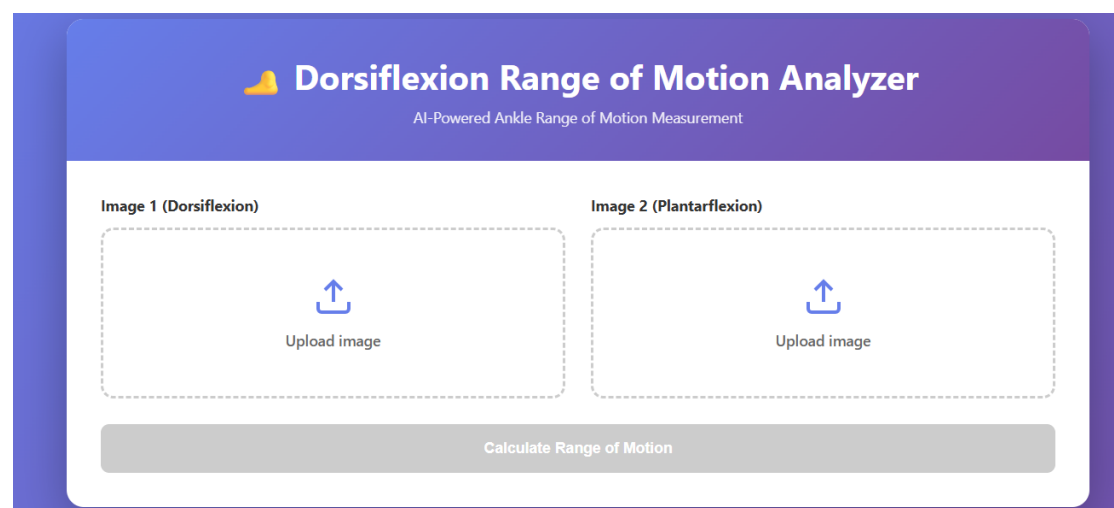


Figure A3: Main screen of the Dorsiflexion Range of Motion Analyzer.

4. Uploading the First Image

To upload the first image, click the “Upload image” area under “Image 1”.

Select the image from the device. After the upload is completed, the selected image will be displayed inside the upload panel.

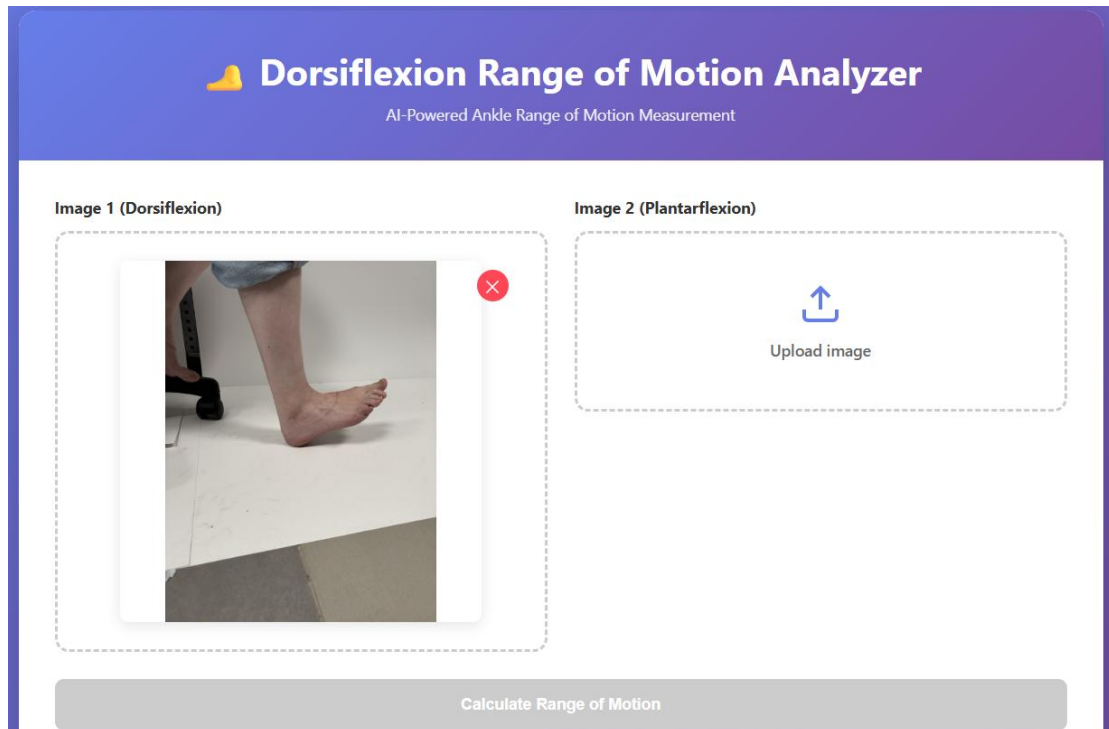


Figure A4:Image 1 (Dorsiflexion)

5. Uploading the Second Image

To upload the second image, click the “Upload image” area under “Image 2”.

Select the second ankle image. The participant position, camera distance, and camera angle should remain as similar as possible to those used for the first image.

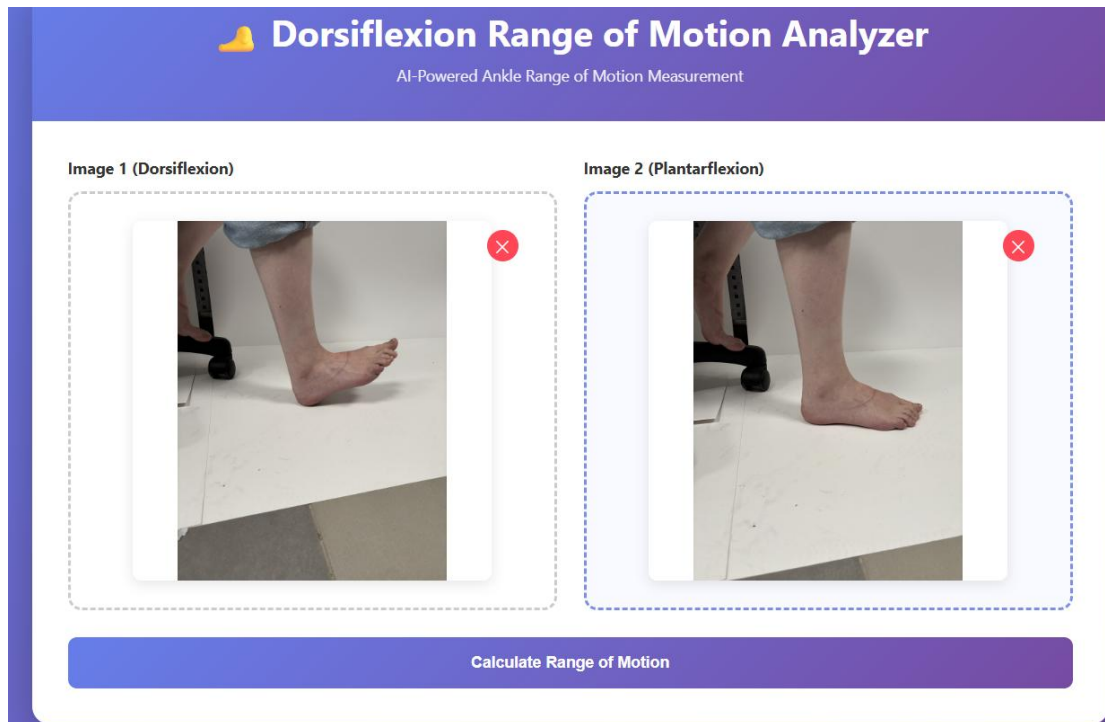
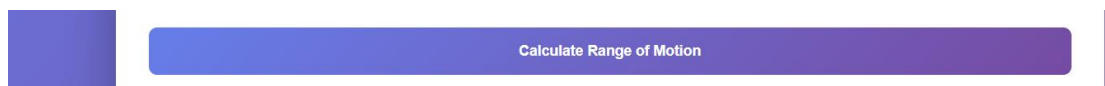


Figure A5:Image 2 (Plantarflexion Position)

6. Calculating the Range of Motion

After both images have been uploaded, click the “Calculate Range of Motion” button.

The system sends the images to the trained model, detects the anatomical landmarks, calculates the ankle angle in each image, and determines the final range-of-motion value.



7. Viewing the Results

After the analysis is completed, the system displays the calculated angle for each image and the final ankle range-of-motion value.

The processed images are shown with the three detected anatomical landmarks:

- Tibial landmark
- Lateral malleolus
- Fifth metatarsal landmark

A confidence score is displayed next to each detected landmark. Confidence values closer to 1 indicate that the model is more certain about the predicted landmark location.

The final ROM value is calculated as the absolute angular difference between the two uploaded images.

If one or more confidence scores are low, the user should verify the landmark positions and consider uploading a clearer image.



Figure A6: Results screen showing the calculated angles, final ROM value, detected landmarks, and confidence scores.

8. System Limitations

The system is currently intended for research and evaluation purposes.

The model was trained on a limited dataset collected under controlled imaging conditions. The results should not be considered a replacement for professional medical diagnosis or an approved clinical measurement procedure.

Programmer Manual

1. System Overview

The Dorsiflexion Range of Motion Analyzer is a web-based application for calculating ankle range of motion from two uploaded images.

The system receives:

- Image 1: Dorsiflexion position
- Image 2: Straight-leg reference position

A trained DeepLabCut model detects three anatomical landmarks in each image:

- Shin landmark
- Ankle landmark
- Fifth-metatarsal landmark

The system then calculates the ankle angle in each image and computes the final range-of-motion value as the absolute angular difference between the two measurements.

The main processing flow is:

Image upload → Image preprocessing → DeepLabCut inference → Confidence validation → Angle calculation → ROM calculation → Result display

2. Project Structure

The main project structure is:

dorsiflexion-ml-clean/

```
|  
|  
|— backend/  
|  |— app.py  
|  
|— frontend/  
|  |— app.js  
|  |— index.html  
|  |— style.css
```

```

|   |— start_server.bat
|   |— start_server.sh
|
|— training_cropped_50/
|   |— config.yaml
|   |— dlc-models-pytorch/
|       |— iteration-0/
|           |— google-pixOct5-trainset80shuffle4/
|               |— test/
|                   |— pose_cfg.yaml
|                   |— train/
|                       |— pytorch_config.yaml
|                       |— snapshot-best-170.pt
|
|— Dockerfile
|— requirements.txt
|— README.md
|— .gitignore
|— .gitattributes

```

Only the files required for running the final system are included in the public GitHub repository. Experimental files, participant images, training images, temporary outputs, and unused model snapshots were omitted.

Main Components

- **backend/app.py** contains the Flask server, API endpoints, image preprocessing, DeepLabCut inference, confidence validation, angle calculation, ROM calculation, and annotated-image generation.
- **backend/uploads/** is created at runtime and stores uploaded and preprocessed images.
- **backend/annotated/** is created at runtime and stores annotated output images.

- **frontend/index.html** defines the user-interface structure.
- **frontend/app.js** handles image upload, API communication, loading states, errors, and result display.
- **frontend/style.css** defines the visual design and responsive layout.
- **training_cropped_50/** contains the active DeepLabCut project and trained model.

The `__pycache__` directory is automatically generated by Python and is not part of the source-code documentation.

3. Technologies Used

The system uses the following technologies:

- **Python 3.10** - backend logic and model execution
- **Flask** - backend web server and REST API
- **Flask-CORS** - frontend-backend communication
- **DeepLabCut 3.0.0rc13** - anatomical landmark detection
- **PyTorch** - deep-learning runtime
- **ResNet-50** - feature-extraction backbone
- **NumPy** - vector and angle calculations
- **Pandas** - reading DeepLabCut prediction files
- **Pillow** - image loading and preprocessing
- **Matplotlib** - generation of annotated result images
- **PyYAML** - reading and updating config.yaml
- **OpenCV Headless** - image-processing support in server environments
- **HTML, CSS, and JavaScript** - frontend implementation
- **Docker** - application packaging and deployment
- **Hugging Face Spaces** - cloud deployment

4. Software Dependencies

The Python dependencies are listed in requirements.txt:

- flask - serves the frontend and exposes the API.
- flask-cors - enables API access from the frontend.
- numpy - performs geometric calculations.
- pandas - reads DeepLabCut .h5 prediction files.
- matplotlib - draws detected landmarks, vectors, and confidence values.
- pillow - corrects image orientation, resizes images, and converts them to PNG.
- pyyaml - updates the DeepLabCut project path.
- opencv-python-headless - supports image-processing operations without a graphical environment.
- torch, torchvision - run the PyTorch-based DeepLabCut model.
- torchaudio - installed as part of the PyTorch environment, although it is not used directly by the current backend.

- deeplabcut==3.0.0rc13 - performs landmark inference.

Install the dependencies using:

```
pip install -r requirements.txt
```

5. Backend Architecture

The backend is implemented in backend/app.py.

It is responsible for:

1. Serving the frontend files.
2. Receiving two uploaded images.
3. Validating the uploaded files.
4. Converting the images to a standardized format.
5. Running DeepLabCut inference.
6. Extracting landmark coordinates and confidence scores.
7. Validating confidence values.
8. Calculating each ankle angle.
9. Calculating the final ROM value.
10. Creating annotated result images.
11. Returning the results to the frontend.

The backend uses Flask and exposes REST API endpoints.

6. Frontend Architecture

The frontend consists of:

frontend/

├── index.html

├── app.js

└── style.css

index.html

Defines:

- Two image-upload panels
- The calculation button
- Loading indicator
- Error-message area
- Angle result cards

- ROM result card
- Annotated-image panels

app.js

Handles:

- Click-based image upload
- Drag-and-drop upload
- File-type validation
- Image preview
- Image removal
- Enabling and disabling the calculation button
- Sending images to the backend
- Displaying loading states
- Displaying errors
- Displaying calculated angles
- Displaying the final ROM value
- Loading annotated images

style.css

Defines:

- Page layout
- Upload panels
- Result cards
- Responsive design
- Loading animation
- Error styling
- Mobile layout

On screens smaller than 768 px, the interface changes from a two-column layout to a single-column layout.

7. Supported Input

The system requires two image files.

The frontend sends the uploaded files using the field names `image1` and `image2`.

Supported file formats:

PNG/JPG/JPEG

The maximum upload size is:

16 MB

The calculation button remains disabled until both images are selected.

8. Image Preprocessing

Before inference, each uploaded image is standardized by the function:

```
convert_upload_to_png(...)
```

The preprocessing steps are:

1. Load the uploaded image.
2. Correct EXIF orientation.
3. Convert the image to RGB.
4. Resize the image while preserving its aspect ratio.
5. Make the larger image dimension equal to 1024 px.
6. Center the resized image on a white 1024×1024 canvas.
7. Save the result as PNG.

This preprocessing avoids cropping and keeps the input format consistent with the model-training setup.

9. DeepLabCut Model Configuration

The active DeepLabCut project is located at:

training_cropped_50/

The main configuration file is:

training_cropped_50/config.yaml

Model Configuration

Task: google-pix

Scorer: r2r

Engine: PyTorch

Project type: Single-subject

Backbone: ResNet-50

Training fraction: 0.8

Augmenter: imgaug

Batch size: 12

p-cutoff: 0.6

Cropping: disabled

The model detects three body parts:

shin

ankle

fifth

The skeleton connections are:

shin → ankle

ankle → fifth

These two connections are later used as the vectors for ankle-angle calculation.

10. Active Model Shuffle

The final public repository includes only the active shuffle 4 model. The backend explicitly performs inference using shuffle=4, so the active model is google-pixOct5-trainset80shuffle4.

The inference call also uses trainingsetindex=0 and processes PNG files. This means that the backend uses the first training fraction defined in the configuration file and analyzes the two preprocessed images as frames.

11. DeepLabCut Inference

For every analysis request, the backend creates a temporary folder with a unique identifier. The two preprocessed images are copied into this folder. Any existing .h5 files in the temporary folder are removed before inference begins.

The backend runs DeepLabCut using deeplabcut.analyze_time_lapse_frames. DeepLabCut processes the two PNG images and creates a prediction .h5 file. The backend reads this file using Pandas.

For each image, the system extracts the x-coordinate, y-coordinate, and likelihood value for the shin, ankle, and fifth landmarks. These values are used for confidence validation and angle calculation.

12. Confidence Thresholds

The application uses different operational confidence thresholds for the three landmarks. The minimum confidence for the shin landmark is 0.2, while the minimum confidence for the ankle and fifth-metatarsal landmarks is 0.4.

An image is considered valid only if all three landmarks pass their respective thresholds. If one or more landmarks are below the threshold, the angle is not used. The backend returns HTTP status 422 and the message “Low confidence detection. Please retake one or both images.”

The operational thresholds used by the application are different from the DeepLabCut p-cutoff of 0.6. The p-cutoff is used during model evaluation, while the 0.2, 0.4, and 0.4 thresholds control whether the application accepts a prediction for angle calculation.

13. Angle-Calculation Algorithm

The angle is calculated using the three predicted landmarks. The shin landmark is treated as the first point, the ankle landmark is treated as the central point, and the fifth-metatarsal landmark is treated as the third point.

Two vectors are created. The first vector points from the ankle to the shin, and the second vector points from the ankle to the fifth-metatarsal landmark. The angle between the vectors is calculated using the dot-product formula.

The cosine of the angle is calculated by dividing the dot product of the vectors by the product of their lengths. The value is clipped to the range from -1 to 1 to prevent numerical errors. The inverse cosine is then calculated, and the result is converted from radians to degrees.

If one of the vectors has zero length, the function returns NaN, because a valid angle cannot be calculated.

14. ROM Calculation

The ankle angle is calculated separately for Image 1 and Image 2. The final range-of-motion value is calculated as the absolute difference between the two angles.

The backend uses the formula $\text{range_of_motion} = \text{abs}(\text{angle1} - \text{angle2})$. The frontend then displays the two individual angles and the final ROM value with one decimal place.

15. Annotated-Image Generation

The backend generates an annotated image for each uploaded image. The annotated image displays the three detected landmarks, the confidence value for each landmark, the two vectors used for angle calculation, and the calculated angle.

The annotated images are created using Matplotlib and saved in the backend/annotated directory. If the landmarks pass the confidence thresholds, the calculated angle is displayed in the image title. If the image does not pass the confidence check, the title displays “Low confidence - angle not used.”

16. API Endpoints

The backend provides three main API endpoints. The /api/health endpoint is used to verify that the server is running and that the DeepLabCut configuration, model directory, and frontend files exist.

The /api/analyze-range endpoint receives the two uploaded images using a POST request. The endpoint validates the images, preprocesses them, performs DeepLabCut inference, checks the confidence scores, calculates the two angles, calculates the final ROM value, and returns the result as JSON.

The /api/annotated/<filename> endpoint returns an annotated image from the backend/annotated directory. The frontend uses this endpoint to display the processed images.

17. API Response

A successful response from /api/analyze-range includes the angle calculated for Image 1, the angle calculated for Image 2, the final range-of-motion value, the generated image filenames, confirmation that the annotated images exist, and the confidence scores for all three landmarks in both images.

If a low-confidence detection occurs, the response includes the validity status and confidence scores for both images. The frontend displays the returned error message to the user.

18. Error Handling

The backend returns a 400 Bad Request response if one or both images are missing, no file is selected, or an unsupported file type is uploaded. The supported formats are PNG, JPG, and JPEG.

The backend returns a 422 Unprocessable Entity response if one or more landmarks do not pass the configured confidence thresholds. The response includes the confidence values and indicates which image failed validation.

A 500 Internal Server Error response is returned if the DeepLabCut configuration file is missing, the model folder is missing, DeepLabCut does not generate a prediction file, or another unexpected exception occurs.

The frontend displays any error message returned by the backend inside the error-message area.

19. Local Execution

The recommended method for running the system locally is to execute `python backend/app.py` from the project root directory. The Flask application starts on host `0.0.0.0` and uses port `7860` by default.

After the server starts, the full application can be opened in a web browser at `http://localhost:7860`. Flask serves both the frontend files and the REST API, so no separate frontend server is required.

20. Optional Frontend Scripts

The frontend directory includes `start_server.bat` for Windows and `start_server.sh` for Linux and macOS. These scripts start a simple Python HTTP server on port `8000`.

However, the current frontend uses relative API paths beginning with `/api`, while the Flask backend runs on port `7860`. Therefore, the standalone frontend server does not automatically communicate with the backend unless an additional proxy or API configuration is added.

For the current implementation, the recommended method is to run `backend/app.py` and access the complete system through port `7860`. The standalone frontend scripts are not required for Docker or Hugging Face deployment.

21. Docker Deployment

The application is packaged using Docker. The Docker image is based on `python:3.10-slim`, and the working directory inside the container is `/app`.

The Dockerfile defines the environment variables `PYTHONUNBUFFERED=1`, `PORT=7860`, and `MPLBACKEND=Agg`. It installs the required operating-system packages, including Git, FFmpeg, and graphical libraries needed by OpenCV and DeepLabCut in a headless environment.

The Dockerfile copies the `requirements.txt` file, installs the Python dependencies, copies the complete project into the container, exposes port `7860`, and starts the application using `python backend/app.py`.

A developer can build the image using `docker build -t dorsiflexion-rom-analyzer .` and run it using `docker run -p 7860:7860 dorsiflexion-rom-analyzer`.

22. Hugging Face Deployment

The system is deployed on Hugging Face Spaces using the Docker SDK. The Space configuration is stored in the `README.md` front matter and defines port `7860` as the application port.

The Hugging Face deployment includes the Flask backend, frontend files, DeepLabCut project, PyTorch model, image-processing functions, angle-calculation logic, and Docker configuration.

GitHub is used for source-code documentation and version control, while Hugging Face Spaces is used for running the deployed web application.

The deployed application is available at:

<https://huggingface.co/spaces/Shelizis12/dorsiflexion-measurement>

23. Model Evaluation Results

The model was evaluated using RMSE, RMSE at p-cutoff, mAP, and mAR. The training RMSE for all predictions was 6.07 pixels, while the test RMSE was 8.05 pixels. After confidence filtering, the training RMSE was 5.27 pixels and the test RMSE was 7.52 pixels.

The training mAP was 97.54%, and the test mAP was 94.48%. The training mAR was 98.41%, and the test mAR was 96.76%.

These results evaluate landmark-detection performance and do not replace clinical validation against manual goniometer measurements.

24. Recommended Tests

Unit Tests

- Angle calculation using known coordinates
- ROM calculation
- Zero-length vector handling
- Confidence-threshold validation
- File-extension validation
- Missing-coordinate handling

Integration Tests

- Uploading two valid images
- Running DeepLabCut inference
- Reading the generated .h5 file
- Returning valid JSON
- Displaying annotated images
- Displaying the final ROM value

Input-Validation Tests

- Missing Image 1
- Missing Image 2
- Unsupported file format
- Corrupted image
- Oversized image
- Low-confidence prediction

Deployment Tests

- Local Flask execution
- Docker build
- Docker container startup
- Hugging Face Space startup
- /api/health response

25. Maintenance Notes

Future developers should document any changes made to the model snapshot, active shuffle, landmark names, confidence thresholds, DeepLabCut configuration, preprocessing dimensions, API response structure, supported file formats, angle-calculation logic, Docker configuration, and Hugging Face deployment settings.

The backend automatically updates the DeepLabCut project path at runtime. This allows the same model project to run on a local computer, inside Docker, or on Hugging Face without manually editing the configuration file.

26. Current Limitations

The current system stores uploaded images and annotated result images locally, and the files are not deleted automatically. The system does not use a database and does not support measurement history, user authentication, or manual landmark correction.

The model was trained on a limited dataset and depends on controlled image-capture conditions. Clinical validation against goniometric reference measurements is still required before the system can be used as an approved clinical measurement tool.

27. Future Development

Future development may include automatic deletion of temporary image files, database integration, measurement-history support, user authentication, manual landmark correction, automatic image-quality validation, automatic posture validation, and expansion of the training dataset.

Additional improvements may include stronger robustness to lighting and camera-angle changes, clinical validation against reference measurements, support for additional ankle movements, and longitudinal patient tracking.

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